



RECENZJA PRACY DYPLOMOWEJ / *DIPLOMA THESIS REVIEW*

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NUMER ALBUMU / <i>STUDENT'S ID NO.</i> :	310284
TYTUŁ PRACY/ <i>THESIS TITLE</i> :	Microstructure-Aware Deep Reinforcement Learning for High-Frequency Trading
TYP PRACY / <i>THESIS TYPE</i> :	☐ PRACA LICENCJACKA / <i>UNDERGRADUATE THESIS</i> ☒ PRACA MAGISTERSKA / <i>GRADUATE THESIS</i>
PROMOTOR / <i>SUPERVISOR</i> :	dr Paweł Sakowski
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RECENZJA PRZYGOTOWANA PRZEZ / <i>REVIEW PREPARED BY</i> :	☐ PROMOTOR / <i>SUPERVISOR</i> ☒ RECENZENT / <i>REVIEWER</i>

WYMAGANIA ODNOŚNIE TRADYCYJNYCH PRAC DYPLOMOWYCH

Prace licencjackie

1. muszą mieć jasno określony cel badawczy oraz powinny zawierać pytanie badawcze lub hipotezę badawczą,
2. muszą wykorzystywać aktualną literaturę naukową krajową lub zagraniczną,
3. muszą wykazywać przygotowanie dyplomanta do prowadzenia badań naukowych,
4. muszą wykorzystywać instrumentarium badawcze prezentowane podczas studiów lub wykraczające poza program,
5. mogą mieć formę pogłębionego referatu (o charakterze opisowym), opracowanego na podstawie literatury lub stanowić prezentację badania własnego, przy czym, jeśli praca oparta jest na literaturze przedmiotu, od autora oczekuje się samodzielności konstrukcji i sposobu interpretacji przedstawionego materiału,
6. w przypadku pracy tradycyjnej, powinny mieć objętość do ok. 50 stron standardowego maszynopisu (100 tys. znaków ze spacjami).

Prace magisterskie

1. muszą mieć jasno określony cel badawczy oraz zawierać (adekwatnie do obranej metody) hipotezę badawczą lub pytanie badawcze,
2. muszą wykorzystywać aktualną literaturę naukową zagraniczną, w odniesieniu do teorii, metody, jak i wyników empirycznych,
3. muszą wykazywać umiejętność samodzielnego prowadzenia badań naukowych przez dyplomanta,
4. muszą wykorzystywać instrumentarium badawcze prezentowane podczas studiów lub wykraczające poza program,
5. muszą stanowić prezentację badania własnego,
6. w przypadku pracy tradycyjnej, powinny mieć objętość do ok. 80 stron standardowego maszynopisu (160 tys. znaków ze spacjami).

REQUIREMENTS FOR TRADITIONAL THESES

Undergraduate theses

1. must have a clearly defined research objective and should include a research question or a research hypothesis,
2. must make use of current national or international scientific literature,
3. must demonstrate the graduate's preparation for research,
4. must make use of research instrumentation presented during the studies or beyond the programme,
5. may take the form of an in-depth paper (of a descriptive character), developed on the basis of literature, or constitute a presentation of own research; if the paper is based on a given literature, the author is expected to be independent in the construction and interpretation of the presented material,
6. traditional theses should be up to approximately 50 pages of standard typescript (100,000 characters with spaces).

Graduate theses

1. must have a clearly defined research objective and include (as appropriate to the method chosen) a research hypothesis or a research question,
2. must make use of current foreign scientific literature, with regard to theory, method as well as empirical results,
3. must demonstrate the graduate's ability to conduct research independently,
4. must make use of research instrumentation presented during studies or beyond the programme,
5. must constitute a presentation of own research,
6. traditional theses should be up to approximately 80 pages of standard typescript (160,000 characters with spaces).

I. MERYTORYCZNA OCENA PRACY / ASSESSMENT OF THE THESIS CONTENT(minimum kwalifikujące / *qualifying minimum*: 23pkt.)

PYTANIE / QUESTION	PUNKTY POINTS
1. MOTYWACJA: Jaki cel (problem, zadanie) został postawiony przed pracą? Czy jego znaczenie zostało uzasadnione? <i>1. MOTIVATION: What goal (task or research problem) has been set in the thesis? Has its significance been justified?</i>	0 – 5
The thesis sets out a clear and relevant goal: a continuous-exposure TD3 agent on limit-order-book features, framed explicitly against bar-based strategies (Section 1.1, p.8-9), with significance assessed against an economic threshold rather than against zero [p.10], which is the right choice at tick frequency. The novelty claim, that this is "a comparatively underexplored design combination" [p.28], would be stronger if anchored to the nearest competing work: a cross-check of the bibliography's 71 entries shows that Nevmyvaka, Feng and Kearns (2006) and Sirignano and Cont (2019) are not cited, and engaging them would let a reader judge the marginal contribution directly.	4
2. PYTANIE BADAWCZE LUB HIPOTEZA BADAWCZA: Czy w pracy zostały postawione pytania badawcze i/lub hipotezy badawcze? Jakież? [Wymogi w tym zakresie zostały podsumowane na stronie 1 formularza recenzji.] <i>2. RESEARCH QUESTION OR HYPOTHESIS: Have research questions and/or research hypotheses been posed in the thesis? If so, which ones? [Please note that the requirements in this matter are summarized on the first page of the review form.]</i>	0 – 5
The central research question is clearly stated [p.9] and operationalised into four numbered hypotheses in Section 1.1.2, which gives Chapter VI a clean organising structure, and Chapter VII returns an explicit verdict on each one. Two points are worth resolving: as printed, Hypothesis 1 asserts both that the edge "survives out of sample" and that "it is eliminated rather than merely eroded by the cost of crossing the spread to trade on it" [p.10] in a single sentence, and none of the four hypotheses is accompanied by a null, an alternative, a significance level or a test statistic. The candid concession that Hypotheses 3 and 4 remain unresolved because "two of the three runs did not train" [p.104] is a transparency that deserves credit.	4
3. METODA: Jaką metodą badawczą posłużono się w pracy? Czy jest ona dostosowana do celu pracy? Czy została prawidłowo zastosowana? a) wybór metody i dostosowanie do celu pracy, b) zakres i poprawność użycia. <i>3. METHOD: What research method was used in the thesis? Does it fit the goal of the thesis? Has it been applied properly?</i> <i>a) choice of method and its fit to the goal of thesis, b) scope and adequacy of use.</i>	a) 0 – 5 b) 0 – 5
The choice of TD3, benchmarked against DDPG, PPO, a random-action control and TWAP/VWAP/buy-and-hold, is a coherent design for a continuous-exposure HFT problem, and the twin-critic, delayed-update and soft-target-update mechanics are correctly specified in Chapter IV. The evaluation would benefit from being built out: every headline result rests on "one completed evaluation run per algorithm for this comparison" [p.84], no statistical inference is reported in Chapter VI although the thesis notes that such testing was "implemented and exercised, but their results are not among the reported findings" [p.153], and no non-RL baseline is trained despite the thesis itself recommending one [p.110]. Table 4's Sharpe, annualised volatility and maximum drawdown are unweighted averages of six per-instrument runs presented as portfolio statistics [p.86], and the normalisation pipeline is described two different ways in Section 5.3.3 [p.72] and Figure 6 [p.149], so both need reconciling. Issues to address in the methodology: 1. Rerun each headline configuration with at least 10 random seeds and report the distribution of every metric, rather than relying on a single run. 2. Add a linear baseline on the same 10 features, as the thesis itself recommends, and report it alongside TD3/DDPG/PPO in Table 4. 3. Replace the six-instrument cross-sectional average in Table 4 with per-instrument statistics plus one explicitly defined pooled-portfolio calculation. 4. Reconcile Section 5.3.3 with Figure 6 and state clearly which statistics transform the validation and test splits.	a) 4 b) 4
4. WIEDZA: Czy i jaką wiedzę nabytą w toku studiów danego stopnia na studiowanym kierunku wykorzystano w pracy? Jak można ocenić stopień przyswojenia wiedzy (podstawowy, zadowalający,	0 – 5

pogłębiony, rozszerzony)? Czy student potrafi łączyć wiedzę z różnych obszarów, korzystać z dorobku różnych dziedzin (jakich?) itd. 4. <i>KNOWLEDGE: If and what kind of knowledge gained during the given degree studied in the field of studies has been used in the thesis? How to rate the level of acquired knowledge (basic, satisfactory, advanced, deepened)? Has student been able to link alternative sources of knowledge, benefit from other disciplines (which ones) etc.?</i>	
Standard deep-RL machinery is competently applied: twin critics with delayed policy updates, target-policy smoothing, Huber-loss critic training and a correctly formalised Differential Sharpe Ratio recursion are properly derived and connected to the implementations. One definitional correction is needed, since the Markov property is stated twice as a property of the optimal action rather than of the transition and reward distribution [p.31, p.47], and the four time horizons in play, the gamma = 0.9 planning horizon [p.64], the median inter-event gap [p.105], the DSR memory [p.56] and the feature-selection windows, should be reconciled explicitly. Computing the update-to-data ratio implied by Table 10, roughly 75,000 gradient updates over 3,000,000 steps, would also be a natural addition given the action saturation reported at [p.104].	4
5. LITERATURA I DANE: Czy dobór literatury i/lub materiału empirycznego jest trafny i aktualny? a) dobór literatury i/lub materiału empirycznego, b) wykorzystanie ich w pracy. 5. <i>LITERATURE AND DATA: Is the selection of literature and/or empirical material accurate and relevant?</i> a) <i>selection of literature and/or empirical material, b) usage in the thesis.</i>	a) 0 – 5 b) 0 – 5
The bibliography, 71 verified entries with no phantom citations, covers the microstructure canon and the RL lineage, though three attributions need correcting: the signed-trade-flow feature is credited to "Lee & Ready (1991)" [p.136] without applying that algorithm, the feature named hft_vpin is not VPIN as the glossary defines it [p.124], and Kyle's lambda is printed with the single-auction coefficient while attributed to the continuous-time model [p.15]. The empirical material is a clear strength: real MBP-10 event data from DataBento for six Nasdaq names, 18,707,348 training events over 25-27 Feb 2026 with validation and test halves of 3,673,436 and 3,673,439 events on 2 Mar 2026, a granularity well beyond a typical bar-based thesis. The main scope limitation is that four trading days, with the entire test confined to half of one session, leave no room for sub-period, regime or walk-forward robustness checks.	a) 4 b) 5
6. WYNIKI: Czy został zrealizowany cel pracy? Czy zweryfikowano hipotezy badawcze? Czy uzyskano odpowiedź na pytania badawcze? Czy praca zawiera błędy, które rzutują na poprawność wyciągniętych wniosków? 6. <i>RESULTS: Has the aim of the thesis been achieved? Have the research hypotheses been verified? Have research questions been answered? Does the thesis have any mistakes which cast on the correctness of driven conclusions?</i>	0 – 5
The headline result, a 304.94% return with Sharpe 5.035 under frictionless mid-price execution (Table 4, p.85), is the thesis's central empirical claim and deserves a reconciliation pass before it is read at face value. Three figures need recomputing or explaining: the Win plus Lose sums differ across policies (TWAP/VWAP 20.27%, TD3 31.87%, DDPG 41.08%, PPO 48.01%) although at zero fee the share of non-zero-return steps should not depend on the policy given that "Roughly 80-90% of consecutive LOB returns are zero" [p.89]; the "21,776 position changes" [p.99] does not reconcile with the 0.53 turnover printed in the same sentence over 3,673,439 events, a gap of roughly a factor of 45; and Figure 4 shows TD3 ending with cumulative reward below zero while ranking near the top on portfolio value in Figure 3 on the same page, which the text does not yet interpret. Extending the bid/ask repricing [p.105] from one instrument into a six-instrument table, using data already in Table 13, would settle whether the edge survives realistic execution. Suggested defense questions: 1. Given that the mid-price is unchanged on 80 to 90 percent of steps at zero fee, could you walk us through how the differing Win+Lose sums across TD3 (31.87%), DDPG (41.08%), PPO (48.01%) and the identical TWAP/VWAP (20.27%) in Table 4 arise? 2. Could you reconcile the "21,776 position changes" figure on page 99 with the 0.53 turnover printed in the same sentence, given that 0.53 times 3,673,439 test steps implies at least roughly 970,000 non-zero position changes? 3. Would it be possible to extend the bid/ask-repriced result from one instrument to all six, using the half-spread and per-step-edge data already in Table 13?	4
7. LOGIKA WYWODU: Czy sposób prowadzenia i opisu wyводу jest prawidłowy? Czy zachowana została logika wyводу? Czy dobrze poprowadzona została narracja? 7. <i>LOGIC OF THESIS: Is the way of conducting and describing arguments correct? Has the logic of arguments been preserved? Was the narrative well conducted?</i>	0 – 5

The thesis follows a coherent arc from motivation through theory, design and implementation to results, and Chapter VII's limitations discussion is more self-critical than most work at this level, which is a genuine strength. The main alignment issue is between chapters: the conclusion states the agent "beats passive benchmarks on this held-out window" [p.103] while Section 6.1.8 states the same comparison supports "the existence half of Hypothesis 1 (a coherent, risk-controlled policy) but not a claim of benchmark-dominating or deployable performance" [p.93], and one consistent formulation should be chosen. A final end-to-end consistency pass against the thesis's own exhibits would catch the remaining cases of this kind.	4
Suma punktów / Total points:	37

II. FORMALNA OCENA PRACY / FORMAL ASSESSMENT OF THE THESIS

(minimum kwalifikujące / *qualifying minimum*: 7pkt.)

1. TYTUŁY: Czy poprawnie i adekwatnie do treści sformułowano tytuł pracy i jej rozdziałów / podrozdziałów? 1. <i>TITLES: Are the title of the thesis and its subchapters formulated correctly and adequately to its content?</i>	0 - 2
The title, "Microstructure-Aware Deep Reinforcement Learning for High-Frequency Trading," accurately signals the method, the input domain and the application area, and the content delivered across Chapters I to VII matches it. Chapter and section titles are specific and descriptive rather than generic, and a dedicated "Thesis Organization and Chapter Overview" section (1.2, p.11 to 12) orients the reader. On title-to-content adequacy this part of the thesis needs no further work.	2
2. KONSTRUKCJA: Czy konstrukcja pracy jest właściwa? 2. <i>CONSTRUCTION: Is the project structured correctly?</i>	0 - 2
The seven-chapter sequence is conventional and logically ordered, and Chapter VI's four subsections map one-to-one onto the four hypotheses stated in Chapter I, which makes the thesis easy to navigate. Two placement points are worth correcting: the feature-selection scoring formulas needed for Section 4.2.5 sit in Appendix J (p.150 to 152) rather than in Chapter IV, and the sentence disclosing the six-fold smaller training budget for the sensitivity sweeps [p.93] currently sits inside a Section 6.1 subsection trained for three million steps [p.84], so it should be relocated to Sections 6.2 to 6.4 where it applies.	2
3: WSTĘP I ZAKOŃCZENIE: Czy wstęp i zakończenie dobrze informują o zamierzeniach Autora(ki), zawartości pracy, użytej metodzie i osiągniętych rezultatach? 3. <i>INTRODUCTION AND CONCLUSIONS: Does the introduction and the concluding chapter clearly state the Author's intentions, thesis contents, the method used and the results achieved?</i>	0 - 2
The Introduction (p.8 to 12) and Conclusions (p.103 to 113) contain every formally required component, including four numbered hypotheses, a chapter overview, an explicit verdict on each hypothesis and a Limitations section covering execution realism, evaluation scope, partial observability and training workflow (Section 7.2, p.104 to 110). Two formulations should be brought into line with the evidence chapter: Hypothesis 1 discloses its own outcome in the Introduction [p.10] rather than framing an open question, and the Conclusion's claim that the agent "beats passive benchmarks on this held-out window" [p.103] is stronger than Section 6.1.8's own wording [p.93].	2
4. ODWOŁANIA: Czy technika prezentacji cudzego dorobku jest poprawna (odsylacze, cytaty, omówienia, bibliografia itp.)? 4. <i>REFERENCES: Are the referencing and quotation techniques applied correctly (references, quotes, discussions, bibliography etc.)?</i>	0 - 2
The bibliography (p.114 to 119) contains 71 entries, all cited at least once, with no phantom entries, which is a solid baseline. Three groups of items need attention: at least eight sources relied on in the text have no bibliography entry, among them the data vendor cited nine times, the Gymnasium API [p.66] and generalized advantage estimation [p.79, p.140]; at least four attributions misrepresent their sources, including the reward-clipping bound credited to "(Mnih et al., 2015)" [p.57] although that paper clips to [-1, 1] rather than the thesis's [-10, 10]; and formatting is uneven, with Moody and Saffell (1998) carrying no journal, volume or page numbers and "van Hasselt" indexed as "Hasselt, H. van," so citations render as "(Hasselt, 2010)".	1
5. EDYCJA: Czy właściwie skonstruowano tabele, wykresy, aneksy statystyczne itp.? Czy edycja nie budzi zastrzeżeń?	0 - 2

5. <i>EDITION: Are charts, graphs, and statistical appendices built appropriately? Is their graphic form acceptable?</i>	
The central exhibits carry the main empirical claims and need a careful editorial pass so a reader can verify those claims from the table or figure alone. Table 4 (p.85) reports Sharpe, annualised volatility and maximum drawdown as unweighted averages of six per-instrument runs, a convention disclosed only in the text [p.86] and not in the legend, and omits a row for buy-and-hold although Section 6.1.1 calls it "the primary economic benchmark" [p.83]. Three figures need fuller labelling (no currency unit on Figure 3's y-axis, no unit or reward scale on Figure 4's, and Figure 5's axis labelled only at negative values although the plotted series are positive, p.90 to p.91), Figure 4 is never interpreted in the surrounding text [p.89], and Figure 6's sequence of boxes [p.149] should clarify how a fit-then-freeze step relates to the online procedure it names.	1
6. <i>JĘZYK: Czy praca została napisana zwięźle, poprawnie z punktu widzenia literackiego, itp.?</i> 6. <i>LANGUAGE: Was the thesis written concisely, correctly from a literary point of view, etc.?</i>	0 - 2
The prose is technically dense but grammatically sound and readable at the sentence level, so the argument is never hard to follow. The most frequent issue is a mixture of US and UK spelling within the same document, for example "normalization" 22 times against "normalisation" 7, and "optimization" 25 against "optimisation" 2, with both spellings of the first pair appearing on p.78 alone; a single find-and-replace pass would resolve it. A run-on sentence at p.67 and an inflection error in the Polish AI-use declaration on the title page, "prezentują własny wkład intelektualny" where the grammar requires "prezentując", are worth a quick copy-edit.	2
7. <i>TERMINOLOGIA: Czy praca została napisana poprawnie pod względem terminologicznym? Czy ewentualne błędy terminologiczne w pracy rzutują na poprawność wyciągniętych wniosków?</i> 7. <i>TERMINOLOGY: Was the thesis written correctly in terms of terminology? Do any terminological errors in the thesis affect the correctness of the conclusions?</i>	0 - 2
Several foundational terms need a precise redefinition, which matters because the contribution rests on exact microstructure and RL terminology. The Markov property is defined twice as a property of the optimal action rather than of the transition and reward distribution [p.31, p.47]; Kyle's lambda is printed with the single-auction formula while attributed to the continuous-time model [p.15, p.16]; and the feature labelled VPIN is implemented without volume bucketing or averaging, so it should either match the definition in the glossary [p.124] or be renamed. Notation would also benefit from standardisation: the symbol phi denotes the actor network for TD3 and DDPG but the value network for PPO [p.62], the object priced by the simulator carries four different names [p.9, p.53, p.70, p.77], and Algorithm 1 states "Load L3 order book dataset" [p.79] for data identified elsewhere as MBP-10.	1
Suma punktów / Total points:	11

III. INNE UWAGI / ADDITIONAL COMMENTS

1. Zakres ewentualnych poprawek pracy (jeśli praca ich wymaga). 1. <i>Scope of thesis corrections (if required).</i>
2. Czy praca została napisana w formie artykułu naukowego? TAK/NIE 2. <i>Information if the thesis was written in the form of scientific paper. YES/NO</i>
☐ TAK / YES ☒ NIE / NO
3. Czy współautorstwo pracy jest uzasadnione (jeśli praca jest pisana przez dwóch studentów). TAK/NIE 3. <i>Information if co-authorship is justified (if the thesis was written by two students). YES/NO</i>
☐ TAK / YES ☐ NIE / NO
4. Konkursy Czy praca powinna zostać zgłoszona do konkursu na najlepszą pracę licencjacką im. W. Baki (jeśli dotyczy)? TAK/NIE Czy praca powinna zostać zgłoszona do konkursu na najlepszą pracę magisterską im. A. Semkowa (jeśli dotyczy)? TAK/NIE 4. <i>Contests</i>

Should the thesis be submitted to the W. Baka competition for the best undergraduate thesis (if applicable)?
YES/NO

Should the thesis be submitted to the A. Semkow competition for the best graduate thesis (if applicable)? YES/NO

☒ TAK / YES

☐ NIE / NO

IV. OCENA PRACY / GRADING SCALE

- ocena merytoryczna / *substantive assessment* 0 – 45 pkt. (min. kwalifikujące / *min. passing score*: ~50%, 23 pkt.);
- ocena formalna / *formal assessment* 0 – 14 pkt. (min. kwalifikujące / *min. passing score*: ~50%, 7 pkt.);
- ocena łączna / *total score* 0 – 59 pkt.

Ocena pracy / <i>Thesis grade</i>	Próg procentowy / <i>Percentage thresholds</i>	Liczba punktów / <i>Points</i>
Ocena celująca / <i>Excellent (5!)</i>	do / <i>up to</i> ~ 100%	57-59
Ocena bardzo dobra / <i>Very good (5)</i>	do / <i>up to</i> ~ 95%	48-56
Ocena dobra i dobra+ / <i>Good and good plus (4, 4+)</i>	do / <i>up to</i> ~ 80%	40-47
Ocena dostateczna i dostateczna+ / <i>Satisfactory, satisfactory plus (3, 3+)</i>	do / <i>up to</i> ~ 65%	30-39
Ocena niedostateczna / <i>Unsatisfactory (2)</i>	do / <i>up to</i> ~ 50%	0-29

Praca wybitna (ocena 5!) – praca znacząco wykraczająca poza program studiów, zawierająca istotny wkład w naukę, nadająca się do publikacji w uznanych czasopismach międzynarodowych.

Outstanding work (grade 5!) - a thesis which significantly exceeds the program of the studies, contains essential input to the science and can be published in recognised scientific journals.

Praca bardzo dobra (ocena 5) – praca w znaczący sposób wykorzystująca treści określone w programie kształcenia, w pełni spełniająca wymagania stawiane pracom dyplomowym.

Very good work (grade 5) - a thesis which significantly uses content specified in the curriculum, which fully meets the general requirements for the thesis.

Praca dobra (ocena 4) – praca w umiarkowany sposób wykorzystująca treści określone w programie kształcenia, w przeważającej części spełniająca wymagania stawiane pracom dyplomowym.

Good work (grade 4) - a thesis which uses content specified in the curriculum in a moderate way and meets the general requirements for thesis in major part.

Praca słaba (ocena 3) – praca w dostateczny sposób wykorzystująca treści określone w programie kształcenia, połowicznie spełniająca wymagania stawiane pracom dyplomowym.

Satisfactory work (grade 3) - thesis sufficiently uses content specified in the curriculum, partly meets the general requirements for the thesis.

Praca zła (ocena 2) – praca w niedostateczny sposób wykorzystująca treści określone w programie kształcenia, w niewielkim stopniu spełniająca wymagania stawiane pracom dyplomowym, zawierająca istotne błędy merytoryczne wpływające na wyciągnięcie niepoprawnych wniosków.

Unsatisfactory work (grade 2) - thesis insufficiently uses content specified in the curriculum, meets the general requirements for thesis in a small extent, contains significant substantive errors which effect in incorrect conclusions.

I. MERYTORYCZNA OCENA PRACY / ASSESSMENT OF THE THESIS CONTENT Suma punktów / total points: (minimum kwalifikujące / <i>qualifying minimum</i> : 23pkt.)	37
II. FORMALNA OCENA PRACY / FORMAL ASSESSMENT OF THE THESIS Suma punktów / total points: (minimum kwalifikujące / <i>qualifying minimum</i> : 7pkt.)	11
Suma punktów / Total points:	48

OCENA / GRADE: Very good (5)

Warszawa, 2026-09-12

(Warsaw, data / date)

Ślepaczuk Robert

(Imię i nazwisko / Name and surname, PODPIS / SIGNATURE)